COMICCRAFT

AI Comic Story Creator

3. Software Requirements Specification (SRS)

Team Information

Team Member	Role
Joseph Thomas	Team Leader
Sugeeve Raj	Team Member
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Functional Requirements

- FR1 – Input Collection: collect story prompt, character name, setting, tone, and art style.
- FR2 – Outline Generation: generate a coherent five-panel outline with title, scene description, and image prompt.
- FR3 – Story Generation: expand the outline with atmospheric captions and engaging narration/dialogue while maintaining continuity.
- FR4 – Image Generation: generate one image for each panel.
- FR5 – Comic Preview: display the generated panels and story information in the browser.
- FR6 – PDF Export: create a PDF containing the comic panels and associated text.
- FR7 – JSON API: expose /generate-comic/json for JSON-based comic generation.
- FR8 – Health Check: expose /health with application status and configuration information.

Input Validation

The PromptRequest model validates story_prompt, character_name, setting, tone, and art_style. The story prompt has a minimum length of 3 and maximum length of 2000 characters; the other fields also have defined limits and whitespace is stripped.

Non-Functional Requirements

- Usability through a simple form-based web interface.
- Maintainability through separate configuration, schema, route, service, template, and test modules.
- Configurability of model IDs, image backend, image dimensions, and storage directories.
- Testability through DEMO_MODE.
- Local execution through Python, Uvicorn, and a web browser.